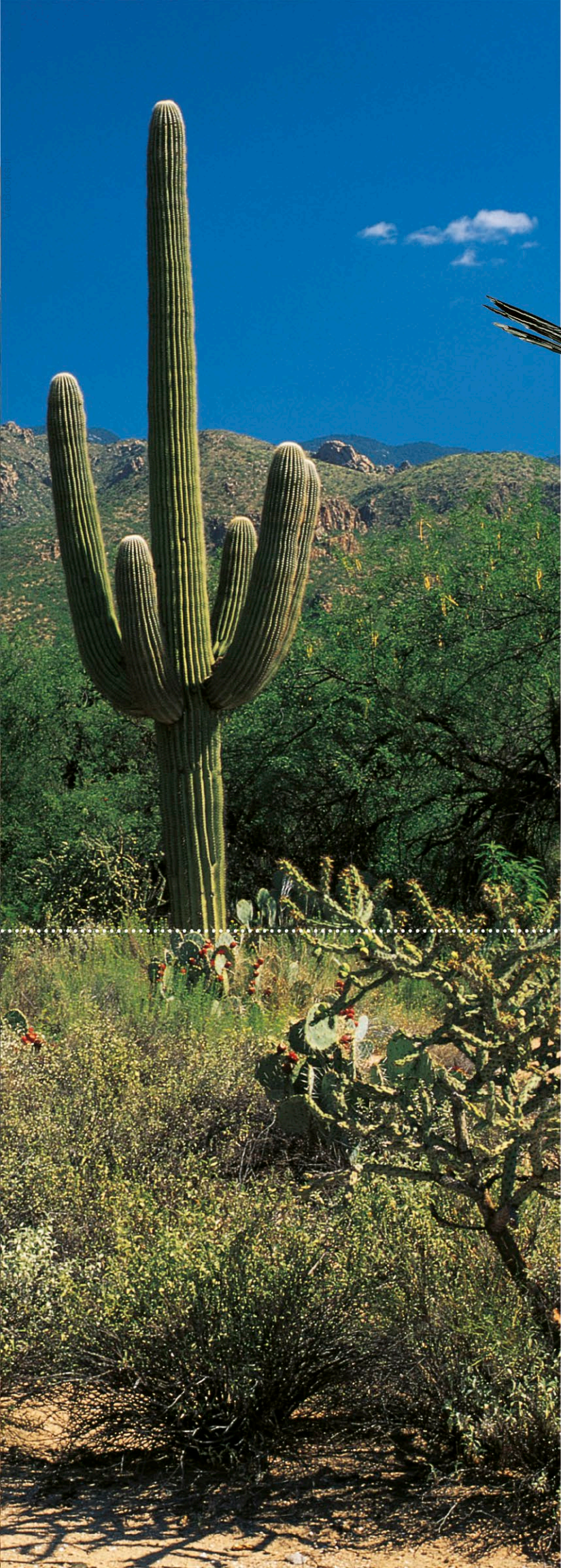
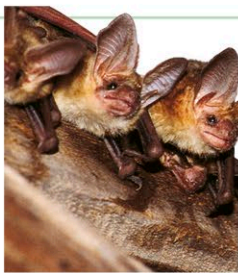




PLANT LAYER



DESERT BATS play a vital ecological role because they feed on insects and pollinate flowers.



SEMIDESERT BIRDS, such as the roadrunner, often nest among spiny plants to protect their young from predators.



THE MEERKAT lives in large colonies. Its varied diet and cooperative foraging technique help it survive when food is scarce.



THE LOCUST'S migratory lifestyle is an adaptation to a habitat where food supplies are erratic and unpredictable.

GROUND



TARANTULAS hunt mainly by touch, enabling them to find their prey after dark.

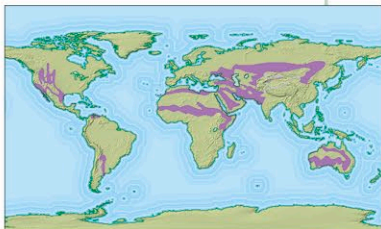
Semidesert

Compared with true desert, semidesert is more widespread, and it is also much more biologically productive. It is found in every continent, including some regions far outside the tropics.

The modest but nevertheless reliable rainfall that semidesert receives has a dramatic effect on the landscape and the types of animals that it can support. Plants often grow in profusion, creating tangled

thickets of vegetation that provide plenty of cover. There are woody species, which store most of their water in underground roots, and fleshy succulents, which store it above ground in their stems and leaves. Most of these desert species are well protected from plant eaters—cacti, for example, have extremely sharp spines, while spurges exude a poisonous milky sap when they are damaged—but, for animals that can overcome these defenses, they are an important source of food. Semidesert also has plants known as desert ephemerals, which spring up rapidly after rain, flower, set seed, and then die. This short life cycle produces extra fresh food for animals, and adds to the stock of seeds scattered over the desert floor.

While some semideserts are warm or hot throughout the year, others are surprisingly cold in winter. In the deserts of central Asia, and in the northern parts of America's Great Basin—the desert region between the Rocky Mountains and the coastal ranges farther west—temperatures can fall to -22°F (-30°C). In these areas, animals need protection against winter cold as well as against summer heat: small animals, such as insects, usually become dormant in winter, and many burrowing mammals hibernate until the spring.



SEMIDESERT is most widespread in the world's major landmasses, and it extends well into the temperate zone in both hemispheres.

DESERTIFICATION

The world's desert regions constantly shift, because rainfall patterns change as time goes by. These changes are generally very slow, which gives wildlife time to adjust. However, deserts can also be created by human activities—particularly poor farming practices, such as overstocking with cattle or goats. This result, called desertification, currently affects many parts of the world, from China and California to the Sahel—the arid region to the south of

the Sahara Desert. Desertification reduces plant cover and speeds up erosion, driving out animals that normally survive in dry habitats. Once it has occurred, it is difficult to reverse.



SAND STORM
Looming high into the sky, a vast cloud of sand approaches a desert town. A powerful storm can displace millions of tons of sand.